

GLOFeagles 2026 Challenge Technical Report

Team Name: CyberCorps

1. Introduction

Glacial Lake Outburst Floods (GLOFs) are hazardous events caused by the sudden release of water from glacial lakes. Early identification of glacial lakes and surrounding environmental conditions is important for disaster monitoring and risk assessment.

This project presents an automated deep learning pipeline for glacier-related feature segmentation and GLOF risk estimation using satellite imagery. The proposed system performs semantic segmentation of glacial environments and generates risk assessments based on detected glacial lakes and surrounding glacier features.

2. Dataset

The dataset provided for the challenge consisted of multispectral satellite imagery along with labeled examples representing different environmental classes.

The classes included:

- Water / Glacial Lake
- Snow Cover
- Cloud Cover
- Terrain Shadow
- Moraine
- Debris Cover

A labeled subset containing 60 annotated samples was used for model training and validation. The dataset was divided into:

- Training Set: 48 images
 - Validation Set: 12 images
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3. Methodology

3.1 Model Architecture

The project uses **DeepLabV3+**, a state-of-the-art semantic segmentation architecture designed for accurate pixel-level classification.

DeepLabV3+ was selected because:

- It captures both local and global contextual information.
- It performs well on remote sensing and segmentation tasks.
- It effectively handles complex object boundaries.

3.2 Input Features

The model was configured to process **8-channel multispectral inputs**.

Using multiple spectral bands provides additional environmental information that is not available in standard RGB imagery and improves the model's ability to distinguish glacier-related features.

3.3 Loss Function

A combined loss function was used:

- Cross Entropy Loss
- Dice Loss

The combination improves class discrimination while addressing class imbalance.

3.4 Training Strategy

The model was trained using:

- Optimizer: AdamW
- Learning Rate: 0.0001
- Batch Size: 8
- Image Size: 512 × 512
- Early Stopping
- Cosine Annealing Learning Rate Scheduler

Early stopping was applied to prevent overfitting and preserve the best-performing model checkpoint.

4. Uncertainty Estimation

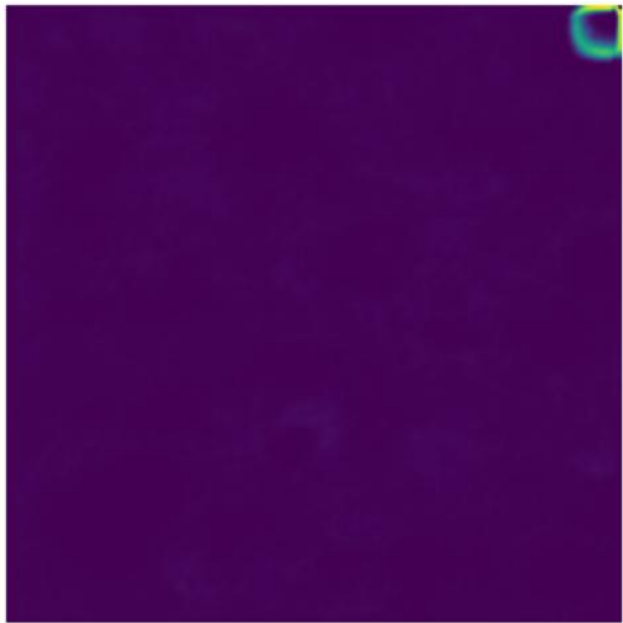
Monte Carlo Dropout was incorporated during inference.

Two stochastic forward passes were used to estimate prediction uncertainty while maintaining computational efficiency. Areas with high uncertainty can be highlighted for additional human review.

This approach improves model interpretability and provides confidence information alongside segmentation outputs.

Figure 1.

Monte Carlo Dropout uncertainty map. Brighter regions indicate higher prediction uncertainty.



5. GLOF Risk Assessment

A risk assessment module was developed to estimate potential GLOF risk levels.

The risk score is calculated using:

- Lake area
- Distance from glacier regions

- Estimated lake growth characteristics

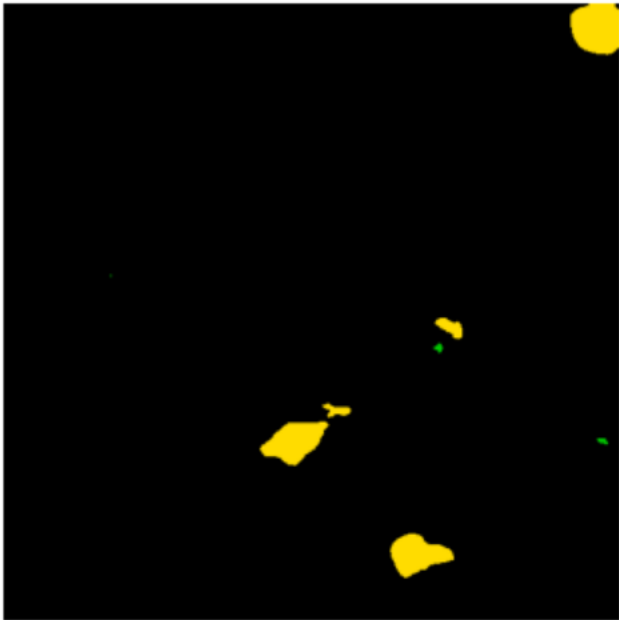
Risk levels are categorized into:

- Low
- Medium
- High
- Critical

Risk maps and JSON reports are automatically generated for each image.

Figure 2.

GLOF risk assessment output generated from segmented lake and glacier features.



6. Results

Validation Performance

Metric	Value
Best mIoU	0.5388

Metric	Value
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Overall Accuracy	95.22%
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The trained model generated predictions for all 575 challenge images using the saved best checkpoint. The best model checkpoint was obtained using early stopping based on validation performance.

Inference Outputs

The final system generated outputs for all 575 challenge images, including:

- Segmentation masks
- Visualization masks
- Uncertainty maps
- Risk maps and per-image risk reports

Figure 3.

Segmentation output generated by the DeepLabV3+ model showing detected glacial lake regions.



7. Design Justification

The following design choices were made:

1. **DeepLabV3+** was selected due to its strong segmentation performance.
2. **Multispectral input channels** were used to capture richer environmental information.
3. **Combined Dice and Cross Entropy Loss** improved segmentation quality.
4. **Monte Carlo Dropout** provided uncertainty estimation.
5. **Risk scoring** transformed segmentation outputs into actionable hazard information.

These decisions were aimed at improving both segmentation accuracy and practical usability for GLOF monitoring.

8. Conclusion

This work demonstrates a complete deep learning pipeline for glacier environment segmentation and GLOF risk estimation. The proposed system successfully generated segmentation outputs and risk assessments for the complete challenge dataset while maintaining reproducibility through saved model checkpoints and source code.